Lecture 6 · Production II: Cost, Wedges, Efficiency

MWG Ch. 5.C–5.D · down the ladder: the rung that survives, and the dispersion puzzle priced

Advanced Microeconomics 1

Thursday 15 October

Why rebuild the theory around cost?

Four reasons, in ascending order of importance:

1. Cost minimization is **necessary** for profit maximization — no extra assumption.
2. It is **well-posed** even under nondecreasing returns, where PMP explodes.
3. It is **where the data are**: firms report expenditures and input quantities, not production sets.
4. It is the ladder's surviving rung: cost minimization only requires price-taking in *input* markets, so it stays valid whatever the firm's conduct in its output market. Today's second half stands on this rung to decompose Monday's dispersion puzzle — and then tests whether even this rung holds.

The cost minimization problem

CMP

CMP (single output)

Given input prices $w \gg 0$ and target output q:

$$c(w, q) = \min_{z \geq 0} \; w \cdot z \quad \text{s.t.} \quad f(z) \geq q.$$

Solution: **conditional factor demand** $z(w, q)$; value: **cost function** $c(w, q)$.

- FOC (interior): $w_\ell = \lambda \, f_\ell(z^*)$, so $\text{MRTS}_{\ell k} = w_\ell / w_k$ — isoquant tangent to isocost.
- The multiplier: $\lambda = \partial c / \partial q =$ **marginal cost**. Keep this; it is the hinge of today's wedge accounting.
- Formally: this *is* the EMP with f playing u and w playing p. Every Ch. 3 result about (e, h) has a twin about (c, z) — next slide is copy-paste with new letters, and that is the point.

Properties of c and z

Proposition · MWG 5.C.2

For f continuous with nonempty upper contour sets:

1. c is **h.d.1 in** w and **nondecreasing in** q; z is **h.d.0 in** w.
2. c is **concave in** w.
3. **Shephard's lemma**: if $z(\bar{w}, q)$ is a singleton, $\nabla_w c(\bar{w}, q) = z(\bar{w}, q)$.
4. If z differentiable: $D_w z = D^2_{ww} c$ is **symmetric, NSD**, with $D_w z\, w = 0$.
5. If f h.d.1 (CRS): c and z are **h.d.1 in** q; if f concave, c is **convex in** q (nondecreasing marginal cost).

- Conditional demands slope down in own input price, symmetrically, *no compensation needed* — output is held fixed, which *is* the compensation.

Trick #3 of 3: the duality trilogy, complete

Concavity of c in w: for fixed feasible $\bar{z}$, $w \mapsto w \cdot \bar{z}$ is linear; $c(\cdot, q)$ is the lower envelope over feasible plans. Lower envelope of linear functions $\Rightarrow$ concave. $\square$

Value function	Problem	Curvature in prices	Envelope: derivative	Derivative matrix
$e(p, u)$	min	concave	$\nabla_p e = h$ (Shephard)	$D_p h$ sym. NSD
$\pi(p)$	max	convex	$\nabla \pi = y$ (Hotelling)	Dy sym. PSD
$c(w, q)$	min	concave	$\nabla_w c = z$ (Shephard)	$D_w z$ sym. NSD

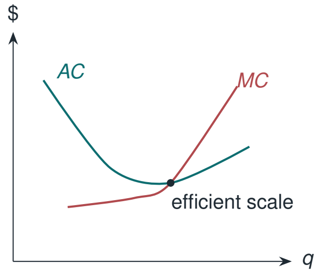
- One theorem, three costumes. If a prelim asks you to prove any row, you now prove all three.
- Empirical payoff of the third row: estimate a flexible $c(w, q)$ (translog), differentiate, and you have the firm's entire input demand system — the workhorse of production econometrics since Christensen–Jorgenson–Lau.

Geometry of cost and supply

From $c(w,q)$ to the cost curves

Fix w; write $C(q) = c(w,q)$, $AC = C(q)/q$, $MC = C'(q)$.

- MC crosses AC at the minimum of AC (differentiate C/q).
- **CRS**: $c(w,q) = q \cdot c(w,1) \Rightarrow AC = MC =$ constant $\bar{c}(w)$; supply is flat at $\bar{c}$, quantity indeterminate — the GE-friendly case.
- Strictly convex C: competitive supply is $p = MC(q)$ above the shutdown threshold ($p \geq \min AVC$ short run; $p \geq \min AC$ for long-run participation).



- Discipline: *fixed* vs. *sunk* is a timing statement, not a synonym pair — fixed costs you still avoid by exit; sunk costs you don't, and only the former belong in the shutdown rule.

Short run vs. long run: LeChatelier

- Short run: some inputs frozen at $\bar{z}_2$; $c^{SR}(w, q; \bar{z}_2) \geq c^{LR}(w, q)$, with equality where $\bar{z}_2$ is optimal — the long-run cost curve is the *lower envelope* of short-run curves. (Envelopes again.)
- **LeChatelier principle** (Samuelson): responses to price changes are weakly *larger in the long run*, when more margins can adjust. E.g. own-price input demand: $\left|\partial z_\ell / \partial w_\ell\right|_{LR} \geq \left|\partial z_\ell / \partial w_\ell\right|_{SR}$.
- Empirical echo: short-run vs. long-run elasticities differ systematically (energy demand, labor demand) — the theory says which way, for free.

Worked example: Cobb–Douglas cost function

$f(z) = z_1^{\alpha} z_2^{\beta}$. Solve the CMP (do it once slowly; the answer is worth memorizing for $\alpha + \beta = 1$):

$$c(w,q) = \gamma\, q^{\frac{1}{\alpha+\beta}}\, w_1^{\frac{\alpha}{\alpha+\beta}}\, w_2^{\frac{\beta}{\alpha+\beta}}, \qquad \gamma = (\alpha+\beta)\,\alpha^{-\frac{\alpha}{\alpha+\beta}}\beta^{-\frac{\beta}{\alpha+\beta}},$$

$$z_1(w,q) = \frac{\alpha}{\alpha+\beta}\cdot\frac{c(w,q)}{w_1}, \qquad z_2(w,q) = \frac{\beta}{\alpha+\beta}\cdot\frac{c(w,q)}{w_2}.$$

- Checks: h.d.1 in w; Shephard ($\partial c/\partial w_1 = z_1$); cost shares constant at $\alpha/(\alpha+\beta)$, $\beta/(\alpha+\beta)$.
- Returns to scale read off the exponent of q: marginal cost is increasing/constant/decreasing as $\alpha + \beta \lesseqgtr 1$.
- Constant cost shares are why Cobb–Douglas production functions and share arithmetic dominate the misallocation accounting — next.

The dispersion puzzle, decomposed

The FOC as a measuring device: wedges

Monday's puzzle: huge within-industry productivity dispersion. The theory's own FOCs turn it into evidence. If firm i maximizes profit facing prices (P, w) *possibly distorted* by a firm-specific wedge $(1+\tau_i)$ (a tax, a credit constraint, a size-dependent rule):

$$(1+\tau_i)\,w \;=\; P\,\frac{\partial F_i}{\partial z} \qquad \Longrightarrow \qquad \mathrm{MRP}_{z,i} \;=\; (1+\tau_i)\,w.$$

The identifying idea (Hsieh–Klenow, QJE 2009)

Undistorted optimization at common prices $\Rightarrow$ **marginal revenue products equalize across firms**. Dispersion in measured MRPs *is* dispersion in wedges — the FOC read backwards, exactly as Afriat read the consumer's.

- Everything on the left is in plant microdata (with Cobb–Douglas, MRPs are revenue-share arithmetic — last slide's dividend). No demand estimation, no conduct assumption: minimal theory, maximal mileage.
- Sanity check: no distortions $\Rightarrow$ MRPs equal $\Rightarrow$ TFPR dispersion zero within industries. The data say: nowhere close.

Goal (b): efficiency — and the rung that cracked

Efficient production

Definition · efficiency (MWG 5.F.1)

$y \in Y$ is **efficient** if there is no $y' \in Y$ with $y' \geq y$ and $y' \neq y$: no plan produces more of something with less of nothing.

Proposition · MWG 5.F.1 — first-welfare flavor

If $y \in y(p)$ for some $p \gg 0$, then y is efficient.

Proof. If y' dominated y, then $p \cdot y' > p \cdot y$ (strictly positive prices), contradicting profit maximization. $\square$

- Two lines, but load-bearing: *any* profit-maximizing plan at strictly positive prices wastes nothing — regardless of convexity, regardless of how peculiar Y is.

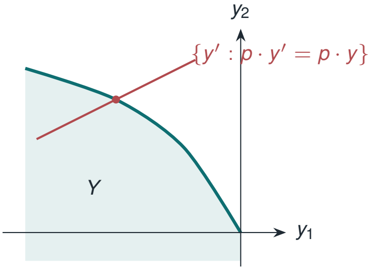
The converse: supporting prices

Proposition · MWG 5.F.2 — second-welfare flavor

If Y is **convex**, every efficient $y \in Y$ is profit-maximizing for some $p \geq 0$, $p \neq 0$.

Proof sketch. y efficient $\Rightarrow$ the open set $\{y' : y' \gg y\}$ misses Y. Both convex $\Rightarrow$ a **separating hyperplane** exists: some $p \neq 0$ with $p \cdot y \geq p \cdot \tilde{y}$ for all $\tilde{y} \in Y$; the monotone side forces $p \geq 0$. $\square$

- Caveats: some prices may be *zero*; and without convexity the theorem **fails** — efficient points inside a nonconvexity have no supporting price.



Goal (b), executed: efficiency made measurable

Application · Hsieh & Klenow (QJE 2009)

- Flip Prop. 5.F.2 into an accounting device: if all firms faced the *same* prices, marginal revenue products would equalize. Firm-specific "wedges" (taxes, credit frictions, size-dependent policy) show up as **dispersion in TFPR** — revenue productivity — within industries.
- Measure that dispersion in plant microdata; compute the TFP gain from reallocating inputs until marginal products equalize (US dispersion as the benchmark): $\approx$ 30–50% in Chinese manufacturing, 40–60% in India.
- "Misallocation": the aggregate cost of *failing to find the supporting prices* — **this block's (b) paper**, and the policy execution: remove the wedges, gain a third to a half of TFP. Care in reading: measurement error inflates dispersion (Bils–Klenow–Ruane 2021); treat magnitudes as upper-bound-flavored.

The bill: money on the table

Application · Bloom, Eifert, Mahajan, McKenzie & Roberts (QJE 2013)

- So far the theory survived intact: firms optimize; the *environment* (wedges) misallocates. The block's third paper tests the optimization rung itself — with a randomized experiment.
- Large Indian textile plants, randomly assigned free management consulting (quality control, inventory, incentives): productivity $+17\%$, profits $\approx +\$300$k/year, within a year.
- The sting: these practices were *known, available, and profitable* — and had not been adopted. Profit maximization says firms do not leave $300k on the table. They did, persistently.

- So the dispersion puzzle decomposes: part environment (wedges — fixable by policy on distortions), part *the firms themselves* (fixable, apparently, by consultants). Both halves are now measured.
- Which makes the next question unavoidable: what *is* the firm maximizing?

What is the firm maximizing? (5.G)

Proposition · unanimity (informal)

If the firm is a price taker and owners' consumption possibilities depend on it only through their share of its *value* $p \cdot y$, then all owners — whatever their preferences — unanimously want profit maximized.

Where it breaks:

- **Market power**: owners who consume the output may prefer lower prices to higher dividends.
- **Incomplete markets**: profit is a random vector; without complete markets to price it, shareholders with different exposures rank plans differently — "maximize profit" is not even well-defined. (A hook for Ch. 6, next week.)
- Modern echo: shareholder-vs-stakeholder debates; Hart–Zingales (2017). The 1995 textbook aside is now a policy fight.

The block's scorecard — and readings

Takeaways

- **Goal (a):** the trilogy complete (e, π, c — one envelope each); the test easy-mode on the top rung, and the surviving rung (cost minimization) turned into a measuring device: MRP dispersion $=$ wedge dispersion.
- **Goal (b), executed:** profit max at $p \gg 0 \Rightarrow$ efficient; the converse needs convexity; and the failure of common supporting prices is priced — 30–50% of TFP in the Hsieh–Klenow counterfactual.
- **The bill:** Bloom et al.'s RCT finds $300k/year lying on factory floors — the optimization rung cracks for organizational reasons, and 5.G stops being a textbook aside.
- Dispersion puzzle, decomposed: part environment, part firm. Both measured; both policy-relevant.

Required: MWG 5.C–5.D, 5.F–5.G.
The block's three papers: *(a)* Syverson (2011); *(b)* Hsieh–Klenow (2009), Secs. I–II; *(limits)* Bloom et al. (2013), intro $+$ Table 1.
In passing: Bils–Klenow–Ruane (2021); Varian (1984); Hart–Zingales (2017).
Monday: Exercise Class 2 (sheet issued today). Uncertainty begins Thursday week: skim MWG 6.A.